

Problem 1

a) Taylor polynomials of $f(x) = x^4 + 2x^3 + x^2 + 5$, $x_0 = 1$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

$$f(x) = x^4 + 2x^3 + x^2 + 5$$

$$f(1) = 9$$

$$f'(x) = 4x^3 + 6x^2 + 2x$$

$$f'(1) = 12$$

$$f''(x) = 12x^2 + 12x + 2$$

$$f''(1) = 26$$

$$f'''(x) = 36x + 12$$

$$f'''(1) = 48$$

$$f^{(4)}(x) = 36$$

$$f^{(4)}(1) = 36$$

$$f^{(5)}(x) = 0$$

$$f^{(5)}(1) = 0$$

$$P_0(x) = f(1) = 9$$

$$P_1(x) = P_0(x) + \overset{12}{f'(1)}(x-1) = 9 + 12(x-1) = 12x - 3$$

$$P_2(x) = P_1(x) + \frac{\overset{26}{f''(1)}}{2}(x-1)^2 = 12x - 3 + 13(x-1)^2 = 13x^2 - 14x + 10$$

$$P_3(x) = P_2(x) + \frac{\overset{48}{f'''(1)}}{6}(x-1)^3 = 13x^2 - 14x + 10$$

b) Taylor series of $g(x) = \ln(1+x)$, $x_0 = 0$

$$g(x) = \sum_{n=0}^{\infty} \frac{g^{(n)}(x_0)}{n!} (x-x_0)^n$$

$$g(x) = \ln(1+x)$$

$$g(0) = 0$$

$$g'(x) = \frac{1}{1+x} \quad \frac{u}{v} = \frac{u'v - uv'}{v^2} \quad g'(0) = 1$$

$$g''(x) = -\frac{1}{(1+x)^2} \quad g''(0) = -1$$

$$g'''(x) = \frac{2(1+x)}{(1+x)^4} = \frac{2}{(1+x)^3} \quad g'''(0) = 2$$

$$\begin{aligned} g(x) &= \overset{0}{g(x)} + \overset{1}{g'(x)}x + \overset{-1}{g''(x)}\frac{x^2}{2} + \overset{2}{g'''(x)}\frac{x^3}{6} \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \end{aligned}$$

Prob 3 - Numerical differentiation

$$u''(x) = \frac{1}{h^2} (a_1 u(x) + a_2 u(x-h) + a_3 u(x+\frac{1}{2}h))$$

Determine a_1, a_2 & a_3

$$u(x-h) = u(x) - u'(x)h + \frac{u''(x)}{2}h^2$$

$$u(x-\frac{1}{2}h) = u(x) - \frac{u'(x)}{2}h + \frac{u''(x)}{8}h^2$$

$$u''(x) = \frac{1}{h^2} (a_1 u(x) + a_2 (u(x) - u'(x)h + \frac{u''(x)}{2}h^2) + a_3 (u(x) - \frac{u'(x)}{2}h + \frac{u''(x)}{8}h^2))$$

$$= \frac{1}{h^2} (u(x)(a_1 + a_2 + a_3) - u'(x)h(a_2 + \frac{1}{2}a_3) + u''(x)h^2(\frac{1}{2}a_2 + \frac{1}{8}a_3))$$

$$\begin{array}{l} u(x) \\ u'(x) \\ u''(x) \end{array} \begin{bmatrix} a_1 & a_2 & a_3 & = u''(x) \\ 1 & 1 & 1 & | 0 \\ 0 & -1 & -\frac{1}{2} & | 0 \\ 0 & \frac{1}{2} & \frac{1}{8} & | 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & | 0 \\ 0 & 2 & 1 & | 0 \\ 0 & 4 & 1 & | 8 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & | 0 \\ 0 & 2 & 1 & | 0 \\ 0 & 1 & 0 & | 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 & 1 & | 0 \\ 0 & 0 & 1 & | -8 \\ 0 & 1 & 0 & | 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & | 4 \\ 0 & 0 & 1 & | -8 \\ 0 & 1 & 0 & | 4 \end{bmatrix}$$

Prob 4 - BVP

$$u_{xx} + 2u_x + \pi^2 u = \cos(\pi x) - \pi(x+1)\sin(\pi x), \quad x \in [0, 2], \quad u(0) = 0, \quad u(2) = 0$$

a) $u(x) = \frac{x}{2} \cos(\pi x)$

$$u_x = \frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x)$$

$$u_{xx} = -\frac{\pi}{2} \sin(\pi x) - (\frac{\pi^2 x}{2} \cos(\pi x) + \frac{\pi}{2} \sin(\pi x)) = -\pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x)$$

$$u_{xx} + 2u_x + \pi^2 u = (-\pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x)) + 2(\frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x)) + \frac{\pi^2 x}{2} \cos(\pi x)$$

$$= \cos(\pi x) - \pi \sin(\pi x) - \pi x \sin(\pi x)$$

$$= \cos(\pi x) - \pi(x+1)\sin(\pi x)$$

b) $u(x+h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} h^n = u(x) + u'(x)h + \frac{u''(x)}{2}h^2$

$$u(x-h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} (-h)^n = u(x) - u'(x)h + \frac{u''(x)}{2}h^2$$

$$u(x+h) + u(x-h) = u(x) + \cancel{u'(x)h} + \frac{u''(x)}{2}h^2 + (u(x) - \cancel{u'(x)h} + \frac{u''(x)}{2}h^2) = 2u(x) + u''(x)h^2$$

$$u''(x) = \frac{u(x+h) + u(x-h) - 2u(x)}{h^2} \Leftrightarrow u''(x) = \frac{U^{i+1} + U^{i-1} - 2U^i}{h^2}$$

central diff: $u(x+h) - u(x-h)$

$$u(x+h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} h^n = u(x) + u'(x)h$$

$$u(x-h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} (-h)^n = u(x) - u'(x)h$$

$$u(x+h) - u(x-h) = \cancel{u(x)} + u'(x)h - (\cancel{u(x)} - u'(x)h)$$

$$u'(x) = \frac{u(x+h) - u(x-h)}{2h} \Leftrightarrow u'(x) = \frac{U^{i+1} - U^{i-1}}{2h}$$

$$u_{xx} + 2u_x + \pi^2 u = \frac{U^{i+1} + U^{i-1} - 2U^i}{h^2} + \frac{U^{i+1} - U^{i-1}}{h} + \pi^2 U^i = f^i = \cos(i\pi) - \pi(i+1)\sin(i\pi)$$

$$F_i = \frac{U^{i+1}}{h^2} + \frac{U^i}{h} + \frac{U^{i-1}}{h^2} - \frac{U^{i-1}}{h} - \frac{2U^i}{h^2} + \alpha^2 U^i$$

$$= U^{i+1} \left(\frac{1}{h^2} + \frac{1}{h} \right) + U^{i-1} \left(\frac{1}{h^2} - \frac{1}{h} \right) + U^i \left(\alpha^2 - \frac{2}{h^2} \right) \quad U^0 = 0, U^N = 1$$

$$F^1 = U^2 \left(\frac{1}{h^2} + \frac{1}{h} \right) + \overset{0}{U^0} \left(\frac{1}{h^2} - \frac{1}{h} \right) + U^1 \left(\alpha^2 - \frac{2}{h^2} \right)$$

$$F^{N-1} = U^N \left(\frac{1}{h^2} + \frac{1}{h} \right) + U^{N-2} \left(\frac{1}{h^2} - \frac{1}{h} \right) + U^{N-1} \left(\alpha^2 - \frac{2}{h^2} \right)$$

$$F^1 = U^2 \left(\frac{1}{h^2} + \frac{1}{h} \right) + U^1 \left(\alpha^2 - \frac{2}{h^2} \right)$$

$$F^{N-1} - \left(\frac{1}{h^2} + \frac{1}{h} \right) = U^{N-2} \left(\frac{1}{h^2} - \frac{1}{h} \right) + U^{N-1} \left(\alpha^2 - \frac{2}{h^2} \right)$$

$$A \vec{x} = b$$

$$\begin{bmatrix} \left(\alpha^2 - \frac{2}{h^2} \right) & \left(\frac{1}{h^2} + \frac{1}{h} \right) & 0 & 0 \\ \left(\frac{1}{h^2} + \frac{1}{h} \right) & \ddots & \left(\frac{1}{h^2} + \frac{1}{h} \right) & 0 \\ 0 & \ddots & \ddots & \ddots \end{bmatrix} \begin{bmatrix} U^1 \\ \vdots \\ U^{N-1} \end{bmatrix} = \begin{bmatrix} F^1 \\ \vdots \\ F^{N-1} \end{bmatrix}$$

Prob 5 - BVP

$$u_{xx} - \frac{\partial}{\partial x} u_x + \frac{\partial^2}{\partial x^2} u = -\frac{x\pi^2}{2} \cos(\pi x), \quad x \in [1, 2] \quad u(1) = -\frac{1}{2} \quad u(2) = 1$$

a) $u(x) = \frac{x}{2} \cos(\pi x)$

$$u_x = \frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x)$$

$$u_{xx} = -\frac{\pi}{2} \sin(\pi x) - \frac{\pi}{2} \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) = -\pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x)$$

$$- \pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) - \frac{\partial}{\partial x} \left(\frac{1}{2} \cos(\pi x) - \frac{\pi x}{2} \sin(\pi x) \right) + \frac{\partial^2}{\partial x^2} \left(\frac{x}{2} \cos(\pi x) \right)$$

$$- \pi \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x) - \frac{1}{x} \cos(\pi x) + \pi \sin(\pi x) + \frac{1}{x} \cos(\pi x)$$

$$\underline{u_{xx} - \frac{\partial}{\partial x} u_x + \frac{\partial^2}{\partial x^2} u = -\frac{x\pi^2}{2} \cos(\pi x)}$$

b) $u_{xx}: u(x+h) + u(x-h)$

$$u(x \pm h) = u(x) \pm u'(x)h + \frac{u''(x)}{2} h^2 \Rightarrow u''(x) = \frac{u(x+h) + u(x-h) - 2u(x)}{h^2} = \frac{U^{i+1} + U^{i-1} - 2U^i}{h^2}$$

$$u_x = u(x+h) - u(x-h)$$

$$u(x \pm h) = u(x) \pm u'(x)h \Rightarrow u''(x) = \frac{u(x+h) - u(x-h)}{2h} = \frac{U^{i+1} - U^{i-1}}{2h}$$

$$u_{xx} - \frac{\partial}{\partial x} u_x + \frac{\partial^2}{\partial x^2} u = \frac{U^{i+1} + U^{i-1} - 2U^i}{h^2} - \frac{\partial}{\partial x} \left(\frac{U^{i+1} - U^{i-1}}{2h} \right) + \frac{\partial^2}{\partial x^2} U^i$$

$$= \frac{U^{i+1}}{h^2} - \frac{U^{i+1}}{xh} + \frac{U^{i-1}}{h^2} + \frac{U^{i-1}}{xh} - \frac{2U^i}{h^2} + \frac{2U^i}{x^2}$$

$$\Delta x = h \Rightarrow x = 1 + ih$$

$$F^i = U^{i-1} \left(\frac{1}{h^2} + \frac{1}{xh} \right) + U^i \left(\frac{2}{x^2} - \frac{2}{h^2} \right) + U^{i+1} \left(\frac{1}{h^2} - \frac{1}{xh} \right), \quad U^1 = -\frac{1}{2} \quad \& \quad U^N = 1, \quad x = 1 + ih$$

$$F^1 = -\frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{xh}\right) + U^1\left(\frac{2}{x^2} - \frac{2}{n^2}\right) + U^2\left(\frac{1}{n^2} - \frac{1}{xh}\right)$$

$$F^1 + \frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{xh}\right) = U^1\left(\frac{2}{x^2} - \frac{2}{n^2}\right) + U^2\left(\frac{1}{n^2} - \frac{1}{xh}\right)$$

$$F^{N-1} = U^{N-2}\left(\frac{1}{n^2} + \frac{1}{xh}\right) + U^{N-1}\left(\frac{2}{x^2} - \frac{2}{n^2}\right) + 1 \cdot \left(\frac{1}{n^2} - \frac{1}{xh}\right)$$

$$F^{N-1} - \left(\frac{1}{n^2} - \frac{1}{xh}\right) = U^{N-2}\left(\frac{1}{n^2} + \frac{1}{xh}\right) + U^{N-1}\left(\frac{2}{x^2} - \frac{2}{n^2}\right)$$

$$Ax = \vec{b}$$

$$\vec{x} = \begin{bmatrix} U^1 \\ \vdots \\ U^{N-2} \\ U^{N-1} \end{bmatrix} \quad b = \begin{bmatrix} F^1 + \frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{xh}\right) \\ \vdots \\ F^{N-1} - \left(\frac{1}{n^2} - \frac{1}{xh}\right) \end{bmatrix}$$

$$A = \begin{bmatrix} \overset{U^1}{\left(\frac{2}{x^2} - \frac{2}{n^2}\right)} & \overset{U^2}{\left(\frac{1}{n^2} - \frac{1}{xh}\right)} & \dots & 0 & 0 \\ 0 & 0 & \left(\frac{1}{n^2} + \frac{1}{xh}\right) & \left(\frac{2}{x^2} - \frac{2}{n^2}\right) \end{bmatrix}$$